

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			0.05 mol (2) award (1) for $\frac{5.23}{106}$ or 0.049 if incorrect answer		2		2	2	2
	(b)	(i)	I	to find out the approximate volume (of acid) needed / to get a rough result so as to save time when carrying out accurate titration	1			1		1
			II	sodium carbonate because less volume of it needed both needed			1	1		1
		(ii)		<u>repeat</u> (the titration) <u>without the indicator</u> (1) 25 cm ³ of Na ₂ CO ₃ (1) 27.65 cm ³ of H ₂ SO ₄ (1) [award (1) for same volume of sodium carbonate <u>and</u> mean volume of sulfuric acid] evaporate the water / leave to evaporate / allow to crystallise (1)	4			4		4
	(c)	(i)		CuSO₄ + 2NaOH 1 mark Cu(OH)₂ + Na₂SO₄ 1 mark 1 mark correct reactants (1) correct products (1) correct balancing (1) – only if reactants and products correct ignore state symbols		3		3		